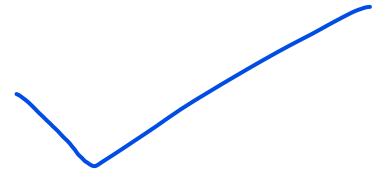


CS-301 Microprocessor Based System Design

Course Teacher: Dr.-Ing. Shehzad Hasan

Lecture - 12

Low Power Features



- Why low power is important?
 - Smaller battery size can result in a smaller sized, lower-cost product
 - It enables products to have a longer battery life
 - It lowers electromagnetic interference (EMI) and improves the quality of wireless communication
 - It simplifies the power supply design, avoiding heat dissipation issues
 - In some instances, a low power embedded system will allow the system to be powered using alternate energy sources (e.g., solar panel, energy harvested from the environment)

Dynamic Energy and Power



- Dynamic energy

- For a single transition i.e. a switch from $0 \rightarrow 1$ or $1 \rightarrow 0$, energy is consumed due to the charging and discharging of capacitances and the switching of transistors

$$\text{Energy}_{\text{dynamic}} \propto \text{Capacitive load} \times \text{Voltage}^2$$

- Power is rate of energy consumption

$$P = \frac{dE}{dt}$$

- Dynamic power

$$\text{Power}_{\text{dynamic}} \propto \text{Capacitive load} \times \text{Voltage}^2 \times \text{Frequency}$$

- For a fixed task, slowing clock rate reduces power but not energy

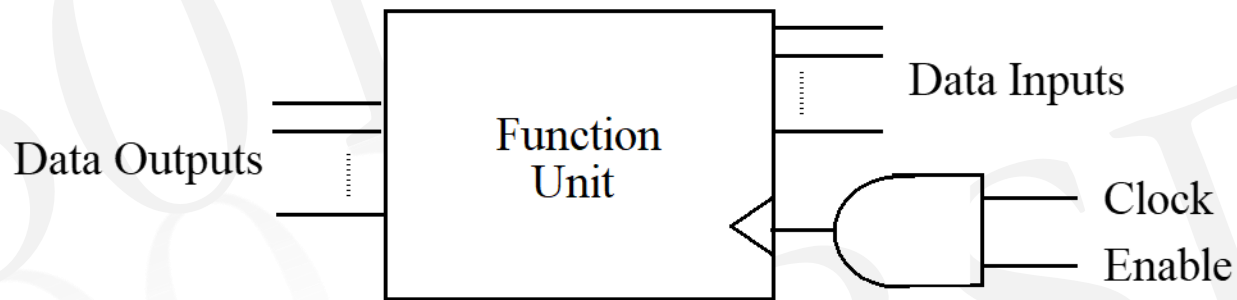
Power Saving Strategies in CMOS

- Reduce Power Supply Voltage (Voltage Scaling)
 - What will be the power saving if supply voltage is reduced from 5V to 3.3V? (assuming frequency is not scaled)
 - 56%
- Lower Clock Frequency (Frequency Scaling)
 - lowers power but not energy consumption (for fixed voltages)
- Power Gating
 - Totally disconnect power from some internal units
 - Saves static power consumption

CMOS = Complementary Metal Oxide
Semiconductor

Power Saving Strategies in CMOS

- Clock Gating: Disable certain internal function units (turn off clock)

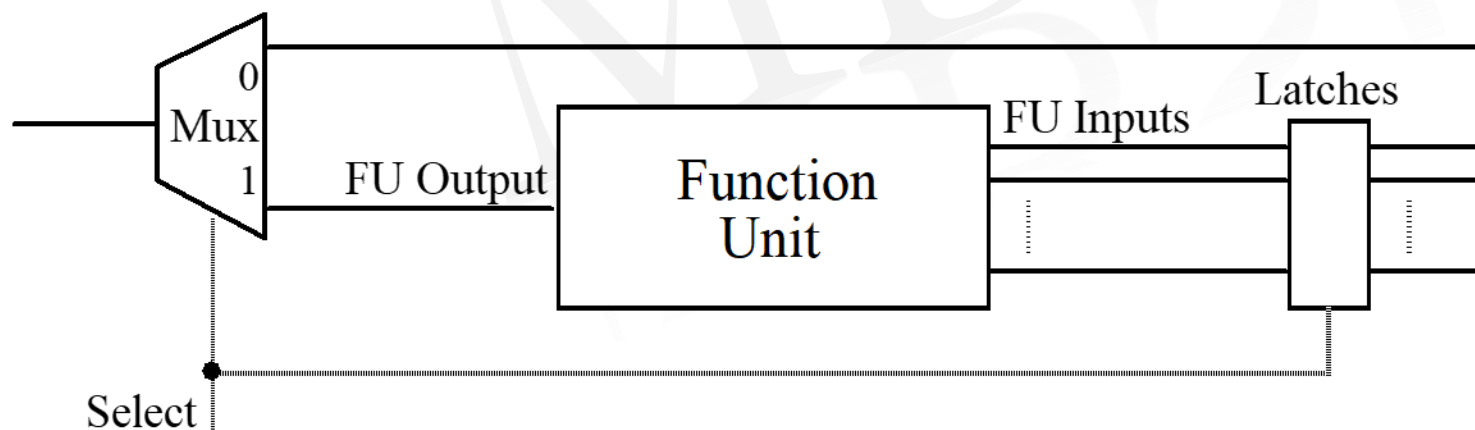


- Guarded Evaluation
 - Functional units are enabled on selected inputs

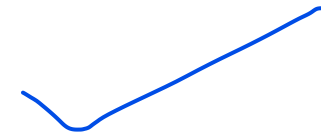
It avoids:

unnecessary computations/switching activity

when results are not needed.



Dynamic Voltage Scaling (DVS)



- This approach exploits the fact that the power consumption of CMOS processors increases quadratically with the supply voltage V_{dd} .

$$P = \alpha C_L V_{dd}^2 f$$

where,

α = switching activity (the fraction of wires that actually toggle)

C_L = load capacitance

V_{dd} = supply voltage

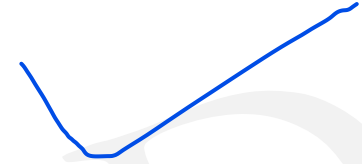
f = operating frequency

Dynamic Voltage and Frequency Scaling (DVFS)

- As voltage is scaled down the propagation delay of CMOS increases linearly

$$\tau = k C_L \frac{V_{dd}}{(V_{dd} - V_t)^2}$$

- Reducing the voltage also means that transistors' switching speed slows down, reducing the maximum frequency at which that circuit can run.
- That is why dynamic voltage scaling is generally done in conjunction with dynamic frequency scaling.
- So a 15% reduction in voltage may result in a 10% reduction in frequency.
 - the exact relationship between how much voltage reduction would lead to how much frequency reduction depends on the system's characteristics and design and the amount of power saving required
 - The main thing is system should function correctly and reliably



Dynamic Voltage and Frequency Scaling (DVFS)

- Example

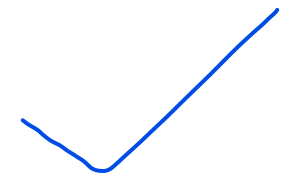
- Assume

- A processor can operate at three different voltages, 2.5 V, 4.0 V, and 5.0 V.
 - Energy consumption is 40 nJ per cycle, at 5.0 V
 - Max. frequency at 5V is 50MHz, voltage reduction leads to linear frequency reduction
 - Our task needs to execute 10^9 cycles within 25 seconds.

- Calculate energy consumption and max. frequency at

- 2.5V and
 - 4V

Dynamic Voltage Scaling (DVFS)



- Example (continued)

Energy $\propto$ Capacitive load $\times$ Voltage²

$V_{dd}(\text{V})$	5.0	4.0	2.5
Energy per Cycle (nJ)	40	25.6	10
$f_{max}(\text{MHz})$	50	40	25
Cycle Time (ns)	20	25	40

40 J

25.6J

$$\frac{Energy_{new}}{Energy_{old}} = \frac{V_{new}^2}{V_{old}^2}$$

Voltage reduction leads to linear frequency reduction

$$c_1 t_1 + c_2 t_2 + c_3 t_3 \leq 25s$$

$$c_1 + c_2 + c_3 = 10^9 \text{ cycles}$$

There are multitude of voltage schedules to execute 10^9 cycles within 25s

Dynamic Voltage Scaling (DVFS)

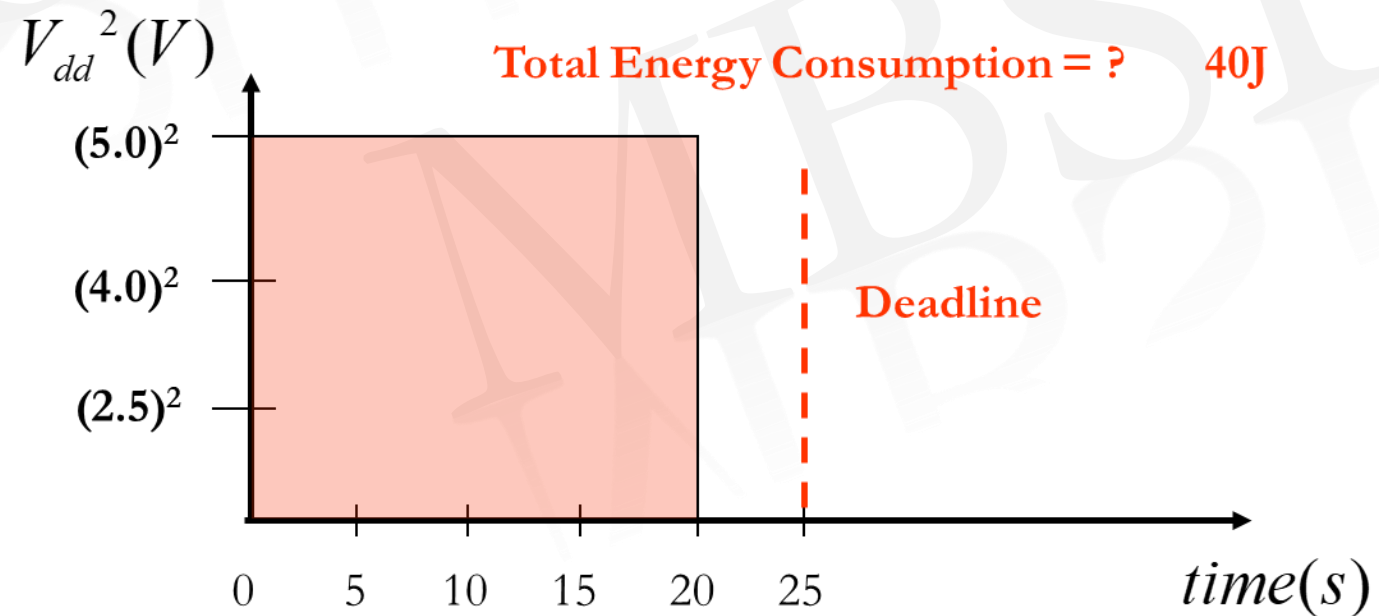
- Example (*continued*)

- Schedule 1

- 50MHz for 20s

10^9 cycles within 25 s

$$time = 10^9 \text{ cycles} \times 20 \times 10^{-9} \text{ s/cycle}$$



Dynamic Voltage Scaling (DVS)

- Example (*continued*)

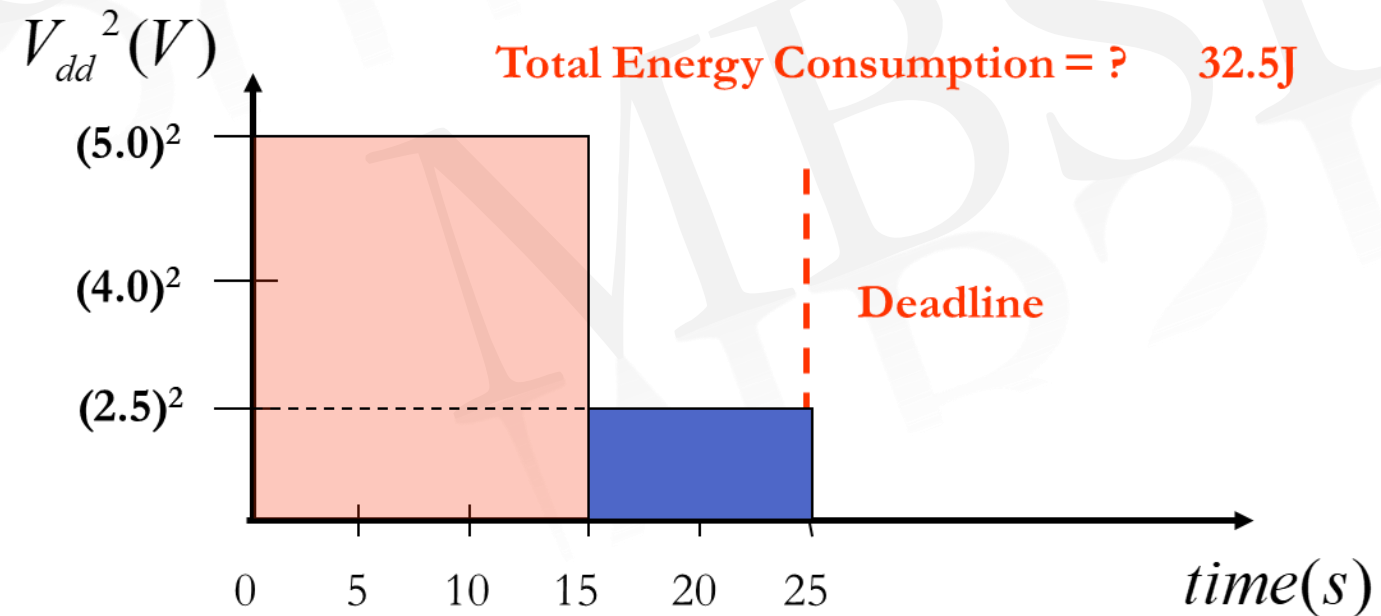
- Schedule 2

- 50MHz for 15s and 25MHz for 10s

10^9 cycles within 25 s

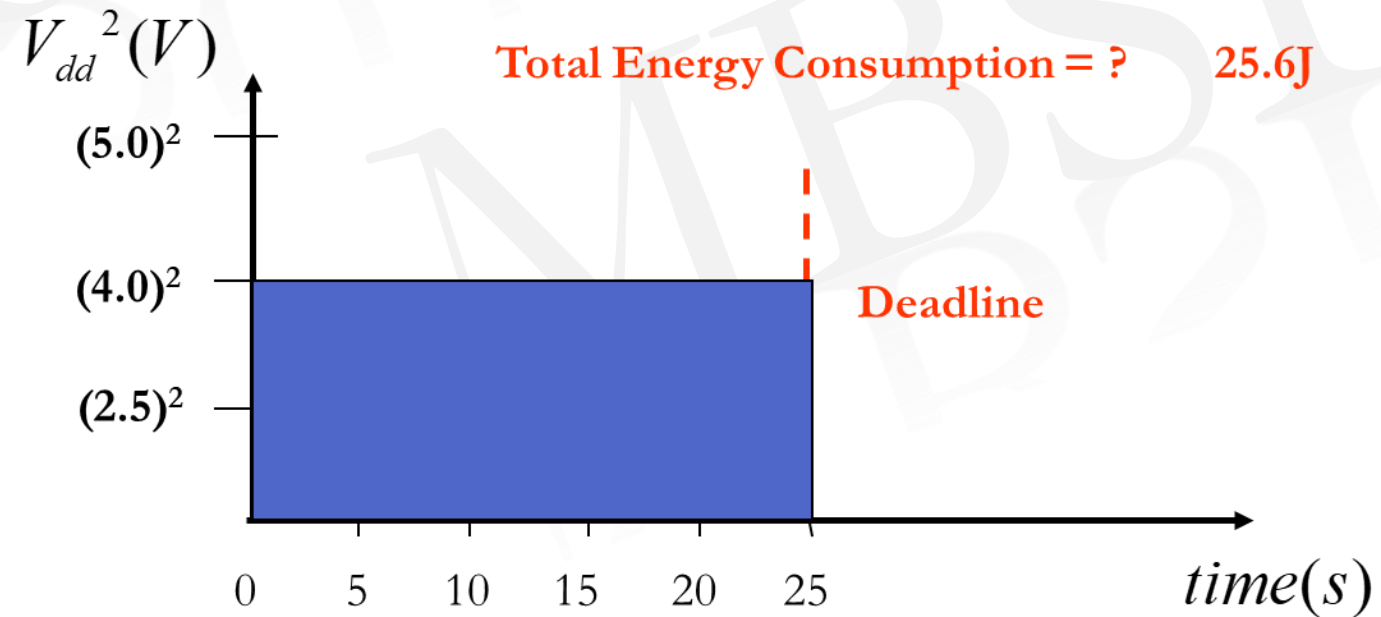
$$c_1 t_1 + c_3 t_3 \leq 25s$$

$$c_1 + c_3 = 10^9 \text{ cycles}$$



Dynamic Voltage Scaling (DVS)

- Example (*continued*)
 - Schedule 3
 - 40MHz for 25s



Observations

- A minimum energy consumption is achieved for the ideal supply voltage of 4 Volts.
- Identify the minimum frequency and therefore closest voltage required to meet the deadline.
 - E.g. if deadline was 30s then 33.33 MHz was the optimal frequency, as this is not available then the combination of the two closest (40 MHz and 25 MHz) would make up the solution
 - Once you have the frequencies you can solve the simultaneous equations to find their times
- It is expensive to support truly variable voltages, and therefore, actual processors support only a few fixed voltages.

CS-301 Microprocessor Based System Design

Course Teacher: Dr.-Ing. Shehzad Hasan

Lecture - 13

Power Management Styles

- Static Power Management

static power does not require activity, it is caused by leakage current and can be reduced by power gating, sleep modes etc

- Does not depend on activity
- Example: user-activated power-down mode.
 - The sensor sends data or interrupts the processor at regular intervals.
 - The processor can be controlled statically to remain in power-gating state for the majority of the time

gating

- Dynamic Power Management (DPM)

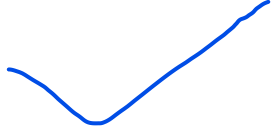
guarded evaluation

- Automatic action based on activity.
- Example: automatically disabling function units.
 - Handling variable bursts of intense processing

Dynamic Power Management (DPM) ✓

- With this approach, processors have several power saving states in addition to the standard operating state.
- Each power saving state has a different power consumption and a different time for transitions into the operating state.

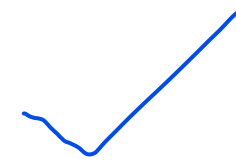
Dynamic Power Management (DPM)

- Goal
 - Energy conservation AND good Performance
 - Need tradeoff between conflicting goals!
 - Fundamental Premises
 - Systems have non-uniform workloads during operation
 - It is possible to predict the fluctuations of workload with some degree of accuracy
 - DPM is performed by a Power Manager
 - Dedicated controllers that monitor activity and orchestrate state transitions
 - Thermal and performance sensors that provide the data needed for the hardware to make power-adjusting decisions
- 

Dynamic Power Management (DPM)

- Problem Formulation

- There are two ways to formulate this problem
 - Minimize power under performance constraints
 - Real-time constraint
 - Optimize performance under power constraints
 - battery lifetime constraint



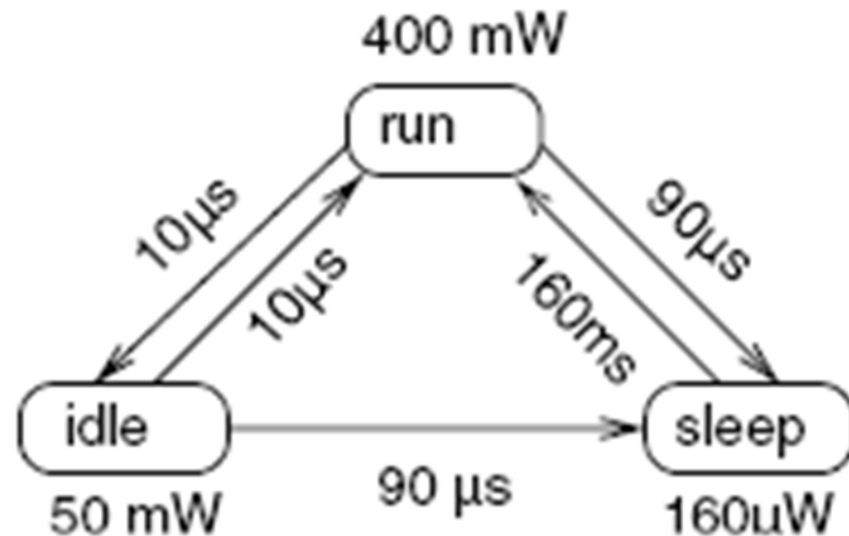
Dynamic Power Management (DPM)

- Power State Machine (PSM)
 - Processor power states are easily modeled by a PSM



Dynamic Power Management (DPM)

- PSM Example



- Run

- This is the normal operation mode and has the highest power consumption

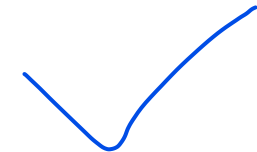
- Idle

- This mode saves power by **stopping** the CPU clock

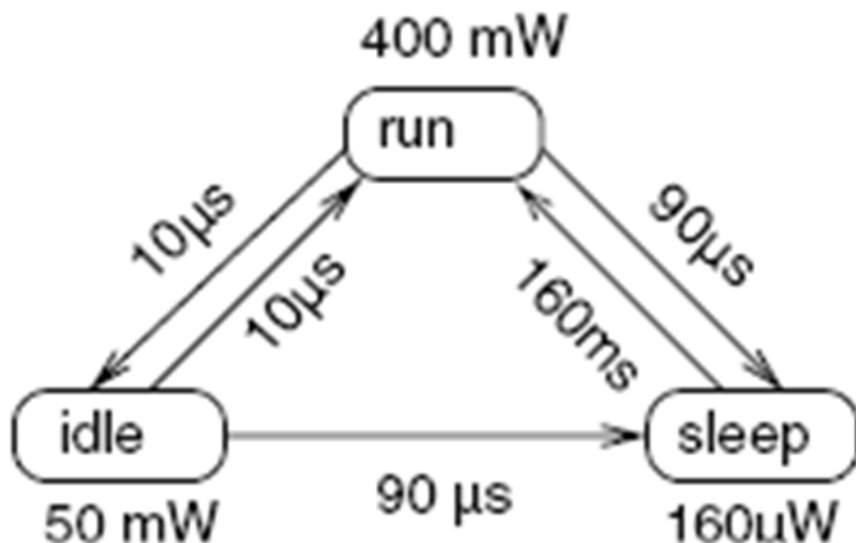
- However, system unit modules – **real-time clock, operating system timer, interrupt control, general-purpose I/O, and power manager** – all remain operational

- CPU returns to run mode upon receiving an interrupt from one of the internal system units or from a peripheral or by resetting the CPU

Dynamic Power Management (DPM)



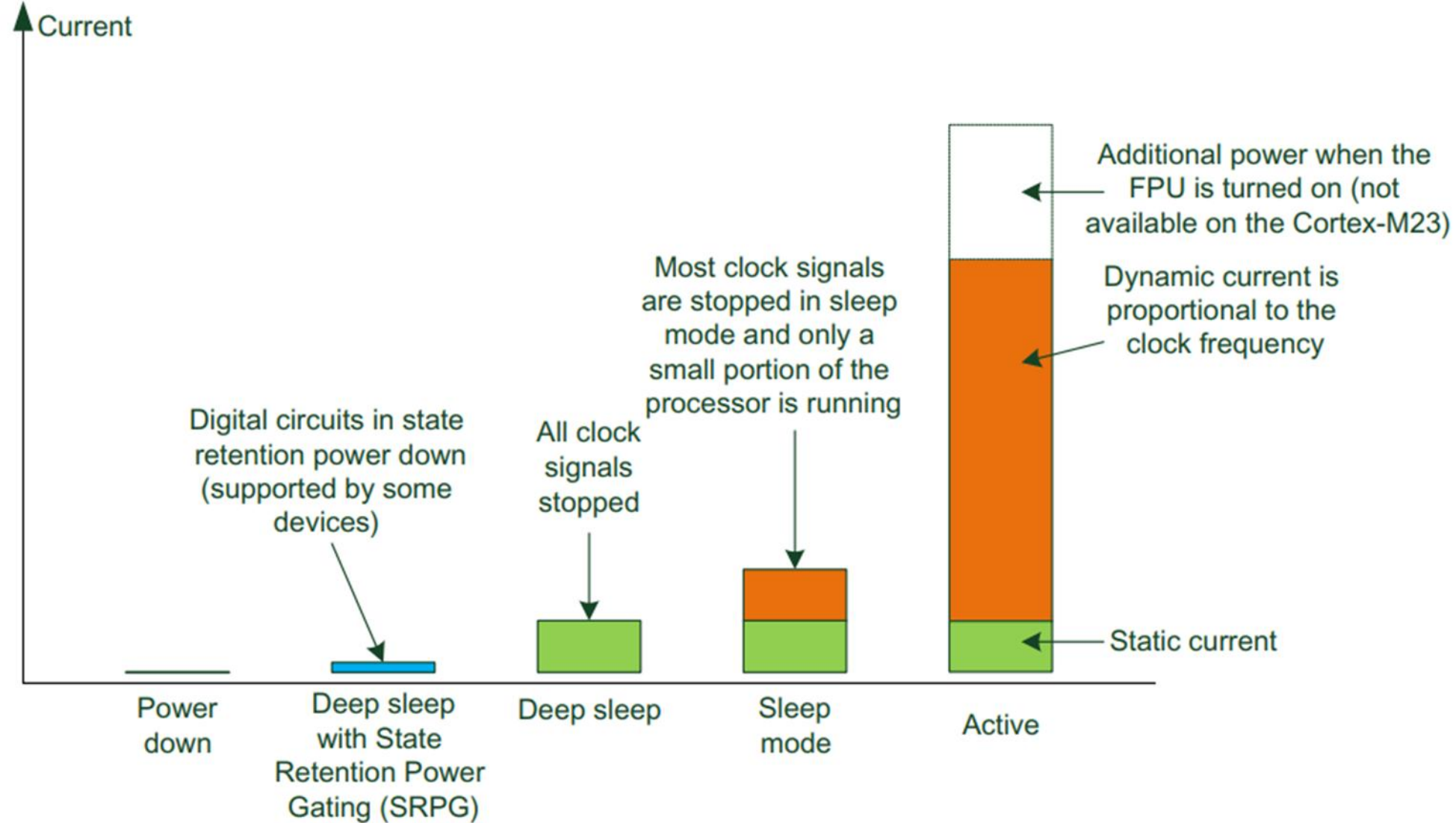
- PSM Example



– Sleep

- causes the system to shut down on-chip activity & reset the CPU.
- a separate I/O power supply remains on and supplies power to the power manager so that the CPU can be awakened from sleep mode.
- The low-speed clock keeps the power manager running at low speeds sufficient to manage sleep mode.

Cortex-M processor power saving modes

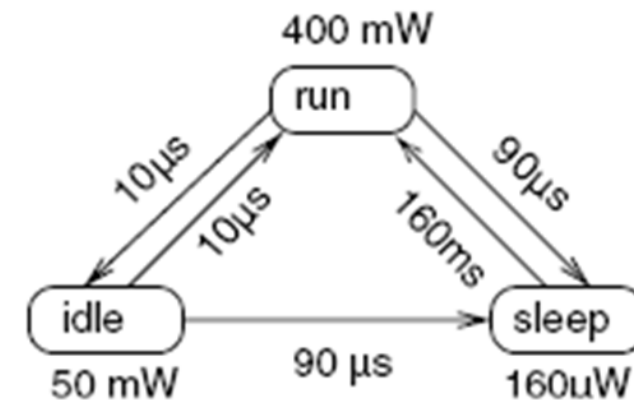


Power Saving Modes – ESP32

- ESP32 offers five power modes
 - Active Mode
 - The processor core, memory, Wi-Fi, Bluetooth, radio, ULP coprocessor, peripherals, RTC are active. (160 – 260 mA)
 - Modem Sleep Mode
 - Everything is active except for Wi-Fi, Bluetooth, radio and peripherals. To keep connections alive they are periodically woken (3 – 20 mA)
 - Light Sleep Mode
 - The CPU core and memory are clock gated (0.8 mA)
 - Deep Sleep Mode
 - CPU core and memory are turned off (power gating) (10 μ A)
 - Hibernation Mode
 - The internal 8 MHz oscillator as well as the ULP-coprocessor are turned off, only the RTC is working (2.5 μ A)



Dynamic Power Management (DPM)



- **Power Down Cost**

- Going into and especially out of a power-down mode is not free – it **costs both time and energy**
 - Power during transitions is same as power consumed in the run state

- **Decision is challenging**

- Avoiding a power-down mode can cost unnecessary power
- Powering down too soon can cause severe performance penalties

Dynamic Power Management (DPM)

- Break-Even Time T_{BE}
 - Minimum idle time required to compensate the cost of entering an inactive state
 - Entering an inactive state is beneficial only if idle time is longer than the break-even time

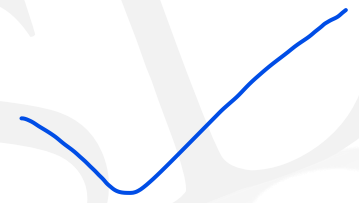
- Assumptions



$$P_{transition} = P_{run}$$

$$T_{BE} = T_{on \rightarrow off} + T_{off \rightarrow on}$$

the system should enter sleep only if $T_{idle} > T_{be}$

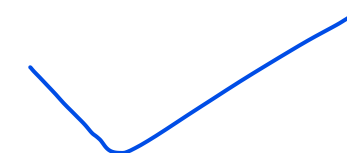


CS-301 Microprocessor Based System Design

Course Teacher: Dr.-Ing. Shehzad Hasan

Lecture - 14

Dynamic Power Management (DPM)



- In most real-world systems, there is little knowledge of future input events and DPM decisions have to be taken based on uncertain predictions.
- All predictive techniques exploit the correlation between the past history of the workload and its near future in order to make reliable predictions.
- We denote by p the future event that we want to predict.
- We denote by o the past event whose occurrence is used to make predictions on p .

Dynamic Power Management (DPM)

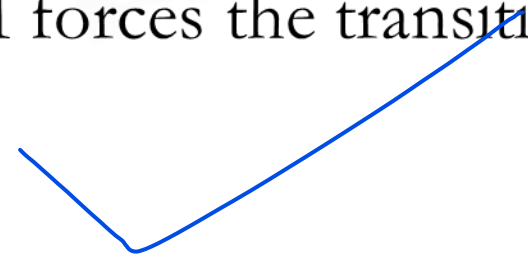
1. Fixed Timeout

- The most common predictive PM policy is the fixed timeout, which uses the elapsed idle time as observed event $o = \{T_{idle} \geq T_{TO}\}$ to be used to predict the total duration of the current idle period

$$p = \{T_{idle} \geq T_{TO} + T_{BE}\}$$

- When an idle period begins, a timer is started with duration T_{TO} . If after T_{TO} the system is still idle, then the PM forces the transition to the Off state.

we didnt imediatey off but wait for TTOperiod



Dynamic Power Management (DPM)

2. Predictive Timeout with Threshold

- Instead of using a fixed timeout observe previous activity i.e. the duration of the busy period immediately preceding the current idle period is observed $\mathbf{o} = \{T_{active}^{n-1} \leq T_{Thr}\}$
- If the busy period is less than some threshold, then the idle period is assumed to be larger than $\mathbf{T_{BE}}$ and the system is transitioned to a low power state.
- The rationale of this policy is that for a class of systems **short active periods are often followed by long idle periods.**

the busy period is the time the cup WAS busy before it become idle

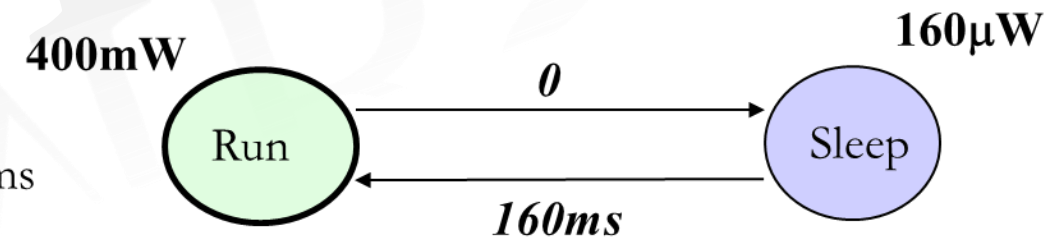
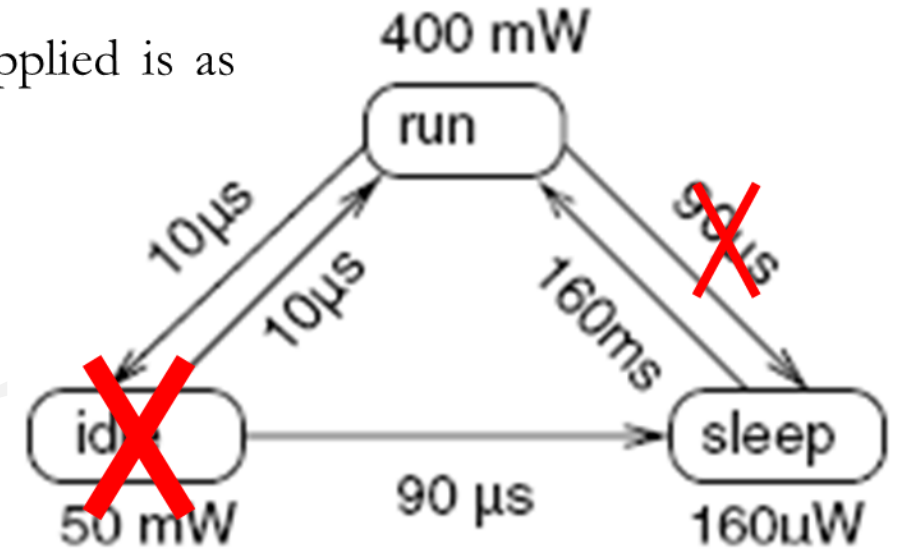
Dynamic Power Management (DPM)

Exercise

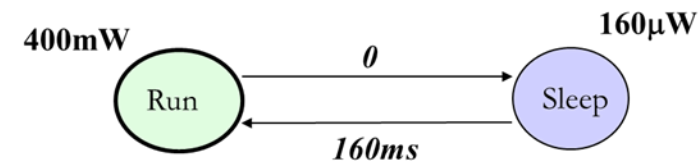
- Suppose a **workload trace** on a processor when DPM is not applied is as follows:

20, 250, 200, 50, 40, 2000, 50, 150, 1500 (ms)

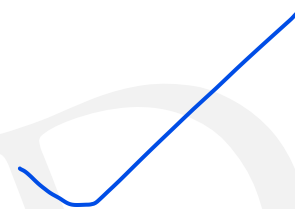
- **Bold red** numbers refer to the duration of an active period.
- We assume the processor starts in the RUN state.
- Only the SLEEP state is used for power saving.
 - The IDLE state is not used.
- For simplicity, we ignore the transition time from RUN to SLEEP.
 - i.e. assume it takes 0 sec.
- Calculate **total run time**, and **energy consumption** of the workload for each of the following DPM policies.
 1. Fixed time-out with a timeout of 60 ms
 2. Threshold-based predictive shutdown with a threshold of 60 ms
 3. Fixed time-out with a timeout of 600 ms



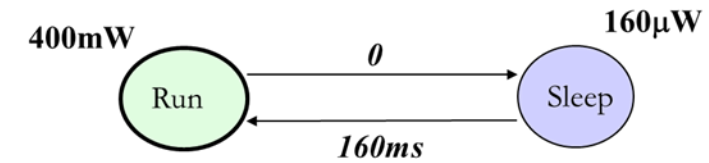
Dynamic Power Management (DPM)



- Fixed time-out with a timeout threshold of 60 ms
- **Solution – 1:** 20, 250, 200, 50, 40, 2000, 50, 150, 1500 (ms)
 - What's the observed event?
 - $\text{idle_time} \geq 60$
 - How many times does this event occur?
 - thrice (250, 2000, 150)
 - For **these idle intervals**, what are the *remaining idle times* after the timeout?
 - 190, 1940, 90



Dynamic Power Management (DPM)



- Fixed time-out with a timeout threshold of 60 ms

- Solution – 1

- Total Execution Time

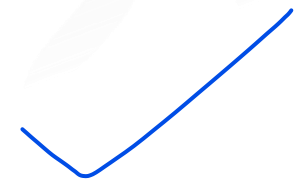
- What's the trace under this policy?

80R 190S 160T 350R 1940S 160T 110R 90S 160T 1500R

20, 250, 200, 50, 40, 2000, 50, 150, 1500

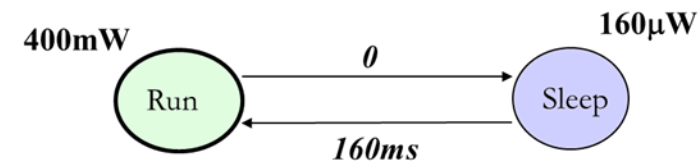
- What's the **total** execution time?

$$\begin{aligned}
 & (80 + 350 + 110 + 1500) + 3(160) + (190 + 1940 + 90) \\
 &= 4740 \text{ ms} \\
 &= 4.74 \text{ s}
 \end{aligned}$$



we have increased the execution time by adding the transition times b the energy consumotion mu st be decreased

Dynamic Power Management (DPM)



- Fixed time-out with a timeout threshold of 60 ms

- Solution – 1

20, 250, 200, 50, 40, 2000, 50, 150, 1500

80R 190S 160T 350R 1940S 160T 110R 90S 160T 1500R

- Energy Consumption

$$(80 + 350 + 110 + 1500) \times P_{\text{RUN}} + (160 + 160 + 160) \times P_{\text{TRAN}}$$

$$+ (190 + 1940 + 90) \times P_{\text{SLEEP}}$$

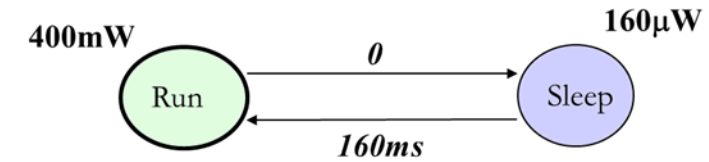
$$= 2040 \times 400 + 480 \times 400 + 2220 \times 0.16$$

$$= 1.0083552 \text{ J}$$

the transition will also be considered as run time so multiplied with run power



Dynamic Power Management (DPM)

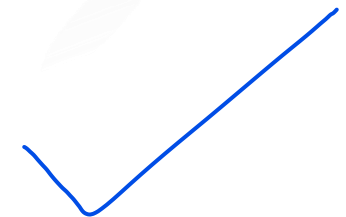


- Threshold-based predictive shutdown with a threshold of 60 ms

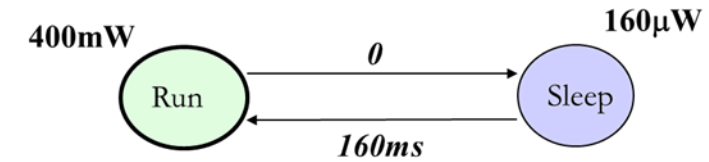
- Solution – 2

20, 250, 200, 50, 40, 2000, 50, 150, 1500

- What's the observed event?
 - » active time ≤ 60
- How many times does this event occur?
 - » thrice (20, 40, 50)
- What are the **idle times** after these active time intervals?
 - » 250, 2000, 150



Dynamic Power Management (DPM)



- Threshold-based predictive shutdown with a threshold of 60 ms

- Solution - 2

- Total Execution Time

- What's the trace under this policy?

20, 250, 200, 50, 40, 2000, 50, 150, 1500

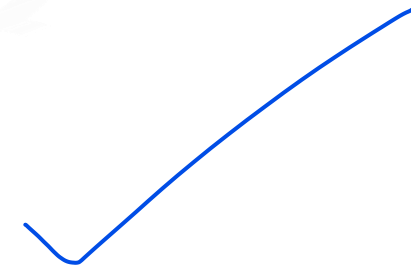
20R 250S 160T 290R 2000S 160T 50R 150S 160T 1500R

- What's the **total** execution time?

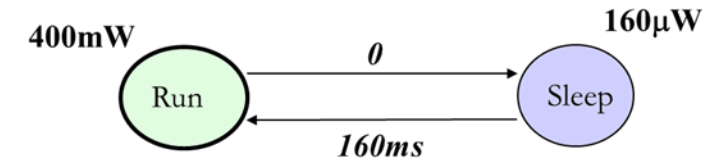
$$(20 + 290 + 50 + 1500) + 3(160) + (250 + 2000 + 150)$$

$$= 4740 \text{ ms}$$

$$= 4.74 \text{ s}$$



Dynamic Power Management (DPM)



- Threshold-based predictive shutdown with a threshold of 60 ms

• Solution

20, 250, 200, 50, 40, 2000, 50, 150, 1500

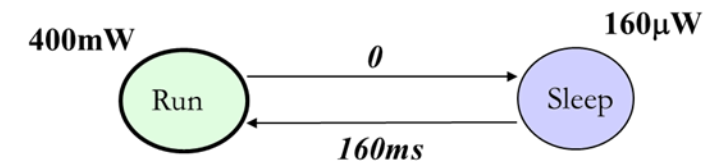
20R 250S 160T 290R 2000S 160T 50R 150S 160T 1500R

• Energy Consumption

$$\begin{aligned}
 & (20 + 290 + 50 + 1500) \times P_{\text{RUN}} + (160 + 160 + 160) \times P_{\text{TRAN}} \\
 & + (250 + 2000 + 150) \times P_{\text{SLEEP}} \\
 & = 1860 \times 400 + 480 \times 400 + 2400 \times 0.16 \\
 & = \mathbf{0.936384 \text{ J}}
 \end{aligned}$$



Dynamic Power Management (DPM)



- Fixed time-out with a timeout threshold of 600 ms

• Solution – 3

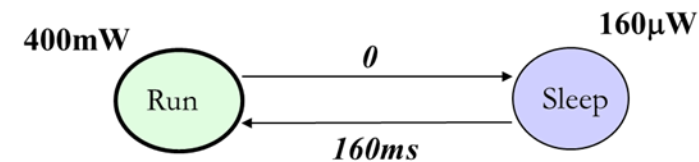
20, 250, 200, 50, 40, 2000, 50, 150, 1500

- What's the observed event?
 - » $\text{idle_time} \geq 600$
- How many times does this event occur?
 - » once – (2000)
- For **this idle interval**, what's the **remaining idle time** after the timeout?
 - » 1400



.

Dynamic Power Management (DPM)



- Fixed time-out with a timeout threshold of 600 ms

• Solution – 3

- Total Execution Time

– What's the trace under this policy?

20, 250, 200, 50, 40, 2000, 50, 150, 1500

$$20+250+200+50+40+600=1160$$

1160R

1400S

160T

1700R

– What's the **total** execution time?

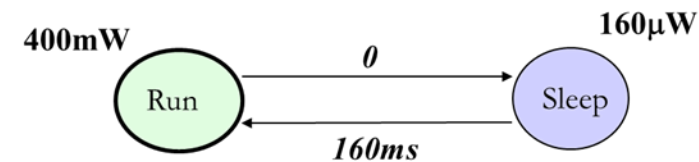
$$(1160 + 1700) + (160) + (1400)$$

$$= 4420 \text{ ms}$$

$$= 4.42 \text{ s}$$



Dynamic Power Management (DPM)



- Fixed time-out with a timeout threshold of 600 ms

20, 250, 200, 50, 40, 2000, 50, 150, 1500

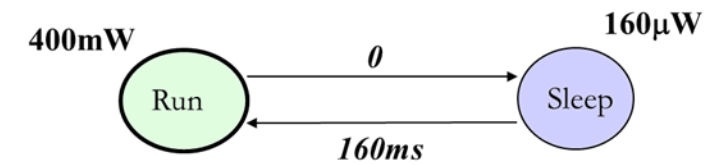
- Solution – 3

- Energy Consumption

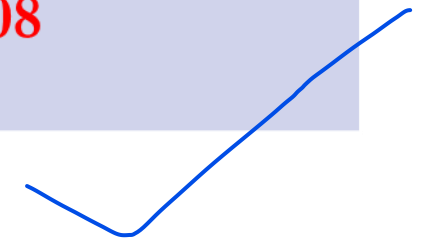
$$(1160 + 1700) \times P_{\text{run}} + (160) \times P_{\text{trans}} + (1400) \times P_{\text{sleep}} = 1.208224 \text{ J}$$



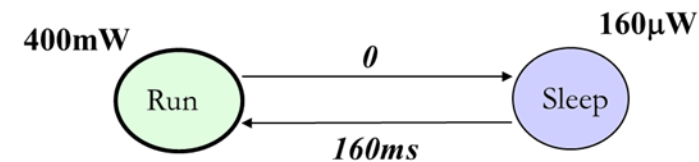
Dynamic Power Management (DPM)



Policy	Time (s)	Energy (J)
Fixed time-out with a timeout of 60 ms	4.74	1.008
Threshold-based predictive shutdown with a threshold of 60 ms	4.74	0.936
Fixed time-out with a timeout of 600 ms	4.42	1.208



Dynamic Power Management (DPM)



- If the workload is performance critical (but less energy constrained), which policy should we use? performance critical = less time less energy constrained = energy can be high
 - Fixed time-out with a timeout threshold of 600 ms
- What if the workload is energy constrained (but less performance critical)?
 - Threshold-based predictive shutdown with a threshold of 60 ms

Dynamic Power Management (DPM)

3. Adaptive Timeout

- Remember in fixed timeout, the elapsed idle time should be greater some fixed threshold $\phi = \{T_{idle} \geq T_{TO}\}$
- What if the timeout value itself changes dynamically according to workload behavior. E.g.
 - increase timeout during bursty workloads
 - decrease timeout during long idle phases
- So we have to look at workload history rather than the last busy period to decide this.
 - during heavy interaction: use $T_{TO} = 200$ ms
 - during quiet periods: use $T_{TO} = 10$ ms
 - during periodic workloads: predict and use $T_{TO} = 3$ ms or immediately sleep

